JIKAI WANG

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RESEARCH INTERESTS

◇ Coupled Human–Environment Systems ◇ Climate Risk Propagation ◇ Climate & Energy Transition Policy

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) *Aug. 2023 – Feb. 2026*
M.S. in Green Growth and Sustainability, Advisor: Prof. Jiyong Eom **GPA:** *3.95/4.30 (96.5/100)*
Relevant Coursework: Energy and Resource Economics (A^+), Climate Policy and Finance (A), Project for Carbon Neutrality Assessments (A), Science of Climate Change (A)

Southwest Jiaotong University (SWJTU), Chengdu, China *Sep. 2019 – Jun. 2023*
B.S. in Statistics, Advisor: Prof. Gaoxiu Qiao **GPA:** *86.21/100*
Relevant Coursework: Econometrics (A), Time Series Analysis (A^+), Multivariate Statistics (A^+), Stochastic Processes (A^-), Real Analysis (A^-)

PEER-REVIEWED PUBLICATIONS

[J4] **Jikai Wang**, Jiyong Eom. Wind power forecast uncertainty co-varies across Korean regions through asymmetric heavy-tailed connectedness networks. ***Communications Earth & Environment***, 2026. [Paper]

[J3] **Jikai Wang**, Gaoxiu Qiao. Extreme events and quantile time-frequency volatility connectedness across crude oil, green bonds and low-carbon equity markets. ***Research in International Business and Finance***, 2025. [Paper]

[J2] **Jikai Wang**, Kai Feng, Gaoxiu Qiao. A hybrid deep learning model for Bitcoin price prediction: data decomposition and feature selection. ***Applied Economics***, 2024. [Paper]

[J1] Yang Hu, Yanran Hong, Kai Feng, **Jikai Wang**. Evaluating the importance of monetary policy uncertainty: the long- and short-term effects and responses. ***Evaluation Review***, 2023. [Paper]

CONFERENCE & WORKSHOP PRESENTATIONS

[W1] **Jikai Wang**. Fragile cooperation: endogenizing climate treaty participation with LLM-driven regional agents. *NeurIPS 2026 Workshop on Tackling Climate Change with Machine Learning*, Dec. 2026. (Poster, Proposals Track) [Workshop]

[C1] **Jikai Wang**, Jiyong Eom. Inherent spatiotemporal uncertainty and co-variation of wind power in Korea. *Korean Operations Research and Management Science Society Fall Conference*, Nov. 2025. (Oral)

RESEARCH EXPERIENCE

Fragile Cooperation: Endogenizing Climate Treaty Participation [W1]
Independent Researcher *May 2026 – Present*
Objective: To study how withdrawal contagion, erosion cycles, and negotiation shape carbon prices and warming.

- Proposed a framework coupling DICE-2023 with regional LLM agents endowed with memory and political identity to endogenize climate-treaty participation.

The Timing of Carbon Taxes over the Business Cycle in a Macrofinancial Agent-Based Model
Advisor: Prof. Jiyong Eom *Dec. 2025 – Present*
Objective: To assess whether adapting carbon taxes to financial conditions can lower transition costs without increasing cumulative emissions.

- Built a two-way coupling between DICE-2023 and the Dystopian Schumpeter meeting Keynes (DSK) macrofinancial agent-based model.
- Designed experiments comparing preset carbon-price paths with tax rules tied to financial conditions, varying policy timing, tax rates, and whether the tax applies at the production or investment stage.

Structure and Sequential Activation of Wind Power Forecast Uncertainty in Korea [J4] [C1]
Master's thesis, KAIST, Advisor: Prof. Jiyong Eom *Nov. 2024 – Dec. 2025*
Objective: To map the spatial structure of wind power forecast uncertainty across its different dimensions, and to simulate the sequence of risk activation following a regional shock.

- Built an hourly forecast error benchmark for Korea's 9 main wind regions and estimated time-varying volatility, skewness, and kurtosis for each region using a GJRSK model. Forecast errors are heavy-tailed across all regions.

- The spatial structure of uncertainty is dimension-dependent. R^2 connectedness networks show that daily volatility concentrates in dominant sources while extreme risks are distributed across multi-polar hubs.
- Applied a Ripple-Spreading Network model to simulate sequential risk activation. Simulated activation followed Korea's northwest-to-southeast monsoon axis and formed two spatial clusters associated with mountainous and plains regions.
- Results are robust across Persistence, ARIMA, and LSTM benchmarks and under winsorization.

Measuring Persistence of Volatility Connectedness under Extreme Events [J3]

Extended undergraduate thesis, Advisor: Prof. Gaoxiu Qiao *Jun. 2023 – May 2025*

Objective: To investigate whether increases in volatility connectedness during extreme events fade afterward or persist as a structural source of systemic risk.

- Estimated TVP-VAR and QVAR time-frequency connectedness among Chinese and international financial markets across 10 main extreme events including COVID-19 and the Russo-Ukrainian war.
- Proposed the Connectedness Survival Rate (CSR) and its quantile extension (QCSR), which measure the share of network edges surviving between adjacent windows, so that connectedness can be assessed by persistence in addition to magnitude.
- Short-term connectedness spikes and fades with each event. Medium- and long-term connectedness form slowly but persist long after the shock. At extreme quantiles the network densifies while its structure turns unstable.
- Chinese green finance markets stayed insulated across market states and serve as hedges. Trading rules built on CSR/QCSR beat single-asset benchmarks on Sharpe ratio, with QCSR reducing maximum drawdown.

Topological Formation and Evolution of Higher-Order Interactions in Social Networks

Remote Research Assistant, National Taiwan University & Lehigh University

Advisor: Prof. Hsuan-Wei Lee (moved from NTU to Lehigh University) *Apr. 2023 – Mar. 2024*

Objective: To investigate how three-person groups form, persist, and dissolve, and what drives these processes.

- Built temporal networks from high school Facebook interaction data and encoded joint interactions among three users as higher-order relations.
- Tracked the emergence, persistence, and dissolution of these structures with persistent homology. Structural shifts clustered around semester starts and holidays.
- Found that low-effort interactions such as likes preceded higher-effort interactions such as tags and posts, suggesting cohesive groups build up gradually.

PROFESSIONAL EXPERIENCE

ENCORED Inc. Seoul, South Korea *Feb. 2026 – Present*

Researcher, Electricity Trading Team, focusing on virtual power plant (VPP) and wind & solar power policy

SELECTED HONORS & AWARDS

KAIST Full Scholarship *2023–2026*

Top 100 Undergraduate Thesis Award, Southwest Jiaotong University *2023*

Outstanding Graduate Award, Southwest Jiaotong University *2023*

Emei 89 Carrying Forward China Scholarship, Southwest Jiaotong University *2022*

One of 20 university-wide recipients selected for academic excellence.

TEACHING & SERVICE

Teaching Assistant, Managerial Economics of Climate & Energy Technologies *Spring 2026*

Teaching Assistant, Scientific Writing *Spring, Fall 2025*

Teaching Assistant, Climate Technopreneurship *Fall 2024*

Administrative Assistant, Department of Green Growth and Sustainability *Spring, Winter 2024*

SKILLS

Methods	*Econometrics*	VAR connectedness, GARCH-family models
		Granger causality (mixed-frequency, extreme, asymmetric)
	Network science	Topological data analysis, ripple-spreading model
	Integrated assessment	DICE-2023, CDICE, JUSTICE
	Agent-based (rule-based)	DSK macro-financial model, Mesa framework
	Agent-based (LLM-driven)	Concordia framework
Programming	Python, R, Julia, MATLAB, LaTeX, C++ (Basic)	
Languages	English (IELTS 7.5), Chinese (Native), Korean (Basic)	